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mixing and distributing humus through out the soil. There are different **endemic** (native) species of earthworms cultured in India for vermicomposting such as *Periyonyx excavatus, Lampito mauritii, Octochaetona serrata.* Some earthworm species have been introduced from other countries and called as **exotic species** Eg. *Eisenia fetida, Eudrilus eugeniae.*

## Vermicomposting

Vermicompost is the compost produced by the action of earthworms in association with all other organisms in the compost unit. Vermicompost bed may be selected on upland or an elevated level as it prevents the stagnation of water. You may construct a cement pit of 3x2x1m size (LxWxD) over ground surface using bricks. The size of pit may vary as per availability of raw materials. Cement pot or well rings are practically good. Provision should be made for excess water to drain. The vermibed should not be exposed to direct sunlight and hence shade may be provided (Figure. 13.1). The first layer of vermibed contains gravel at about 5 cm in height, followed by coarse sand to a thickness of 3.5 cm, which will facilitate the drainage of excess water.

Earthworms collected from native soil prefer a layer of local soil in their compost beds. If local soil earthworms are used, add a layer of native loamy soil for about 15 cm on top of the gravel sand layer and introduce earthworms into it. For exotic species such as *Eisenia fetida* and *Eudrilus eugeniae,* the layer of soil is not needed. The unit can now be loaded with digested biomass or animal dung such as cow dung that has lost its heat. The number of earthworms to be introduced in an unit depends on the size of the vermibed prepared. Earthworms such as *Periyonyx. excavatus, Eisenia fetida* or *Eudrilus eugeniae* are introduced on the top. Jute bags or cardboards or broad leaves are used to cover the unit. As worms require moisture, water management is most important for the survival of the earthworms. Too little or too much of water is not good for the worms.

Earthworms release their castings on the surface. One can start harvesting this from the surface on noticing the castings on the surface. It may take several days for the entire biomass to be composted depending on the amount of biomass.




**Figure 13.1** Vermiculture unit and Earthworms